

Visualizing the Energy Grid

Introduction

This was a project to graph the energy grid and the emission sources occurring on and off of the grid

Goal: Find correlations between greenhouse gas emissions and power infrastructure

Goal: Learn how to use Python and mapping libraries to create detailed, useful data visualizations

Data Used

HIFLD (Homeland Infrastructure Foundation-Level Data)

EPA FLIGHT (Facility-Level Information on Greenhouse Gases Tool)

OpenStreetMap API

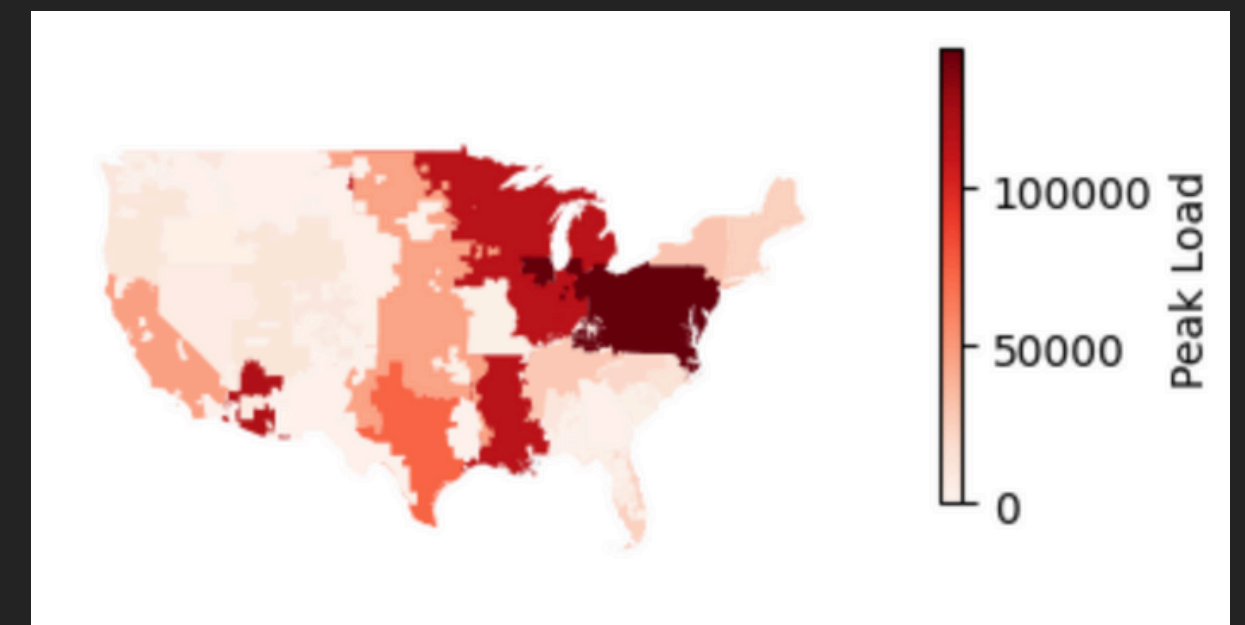
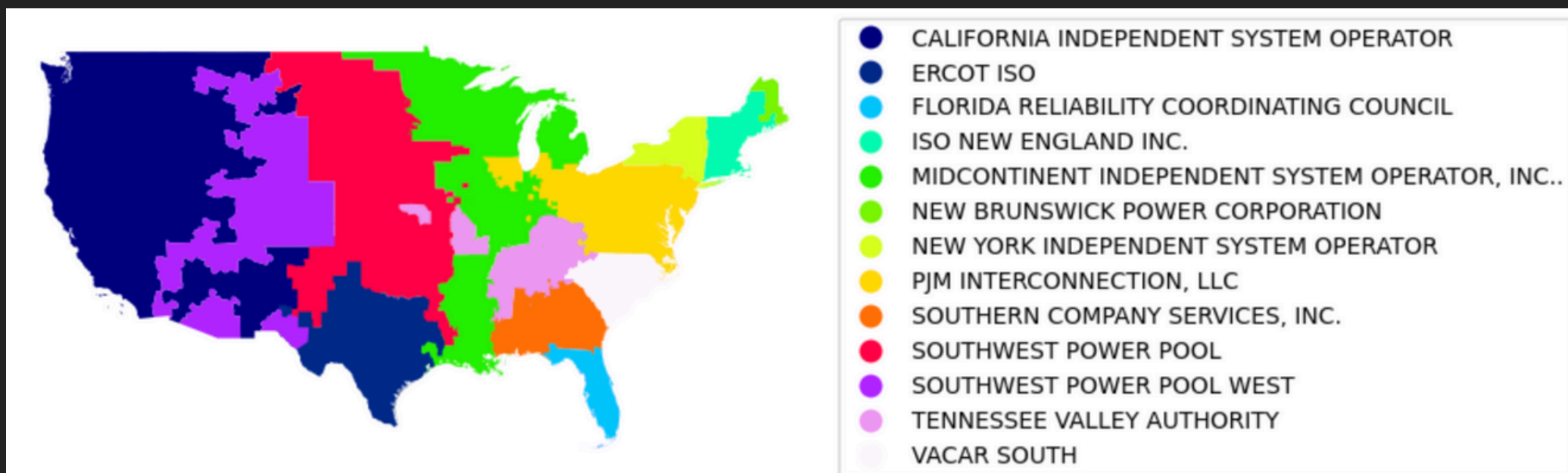
eGRID (Emissions & Generation Resource Integrated Database)

Geopandas

- Tool for visualizing and overlaying data
- Can create plot and choropleth maps
- Allows for spatial data to be overlaid

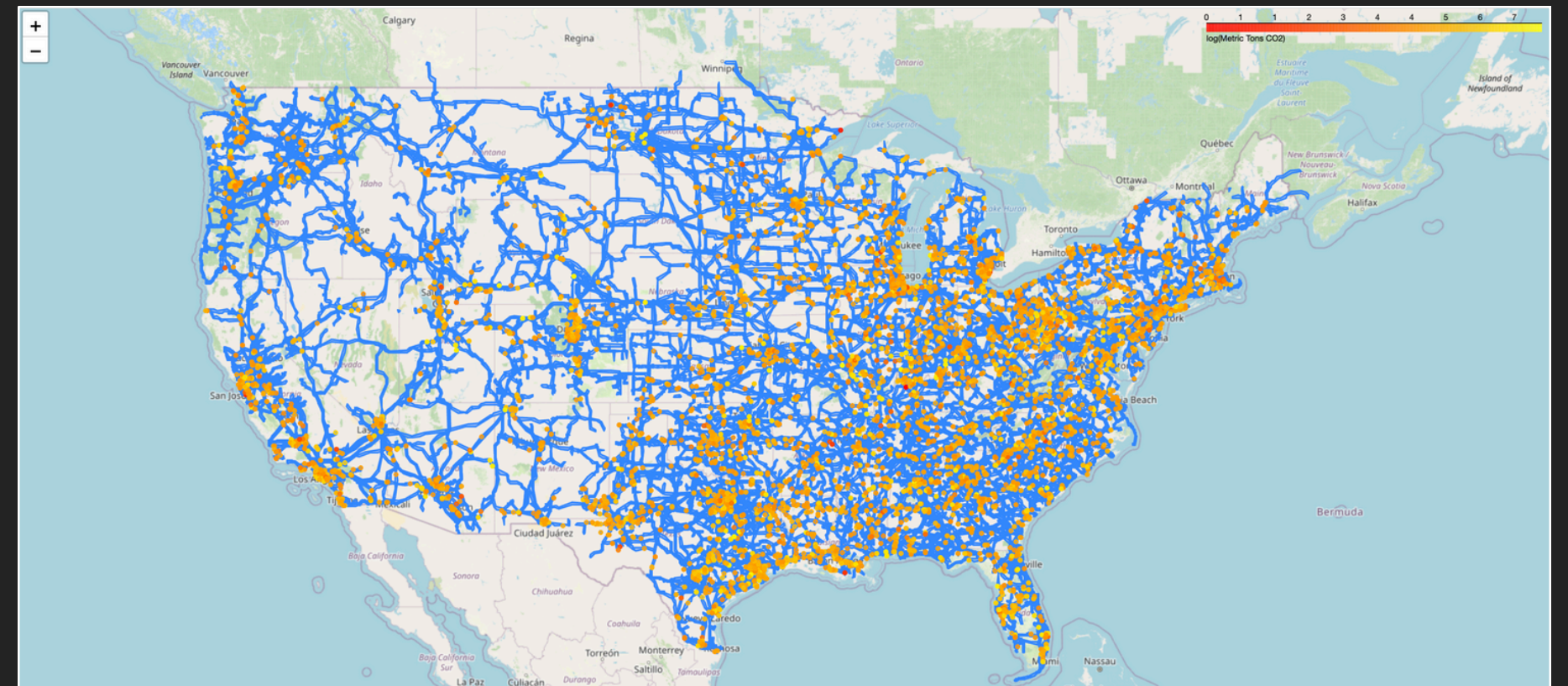
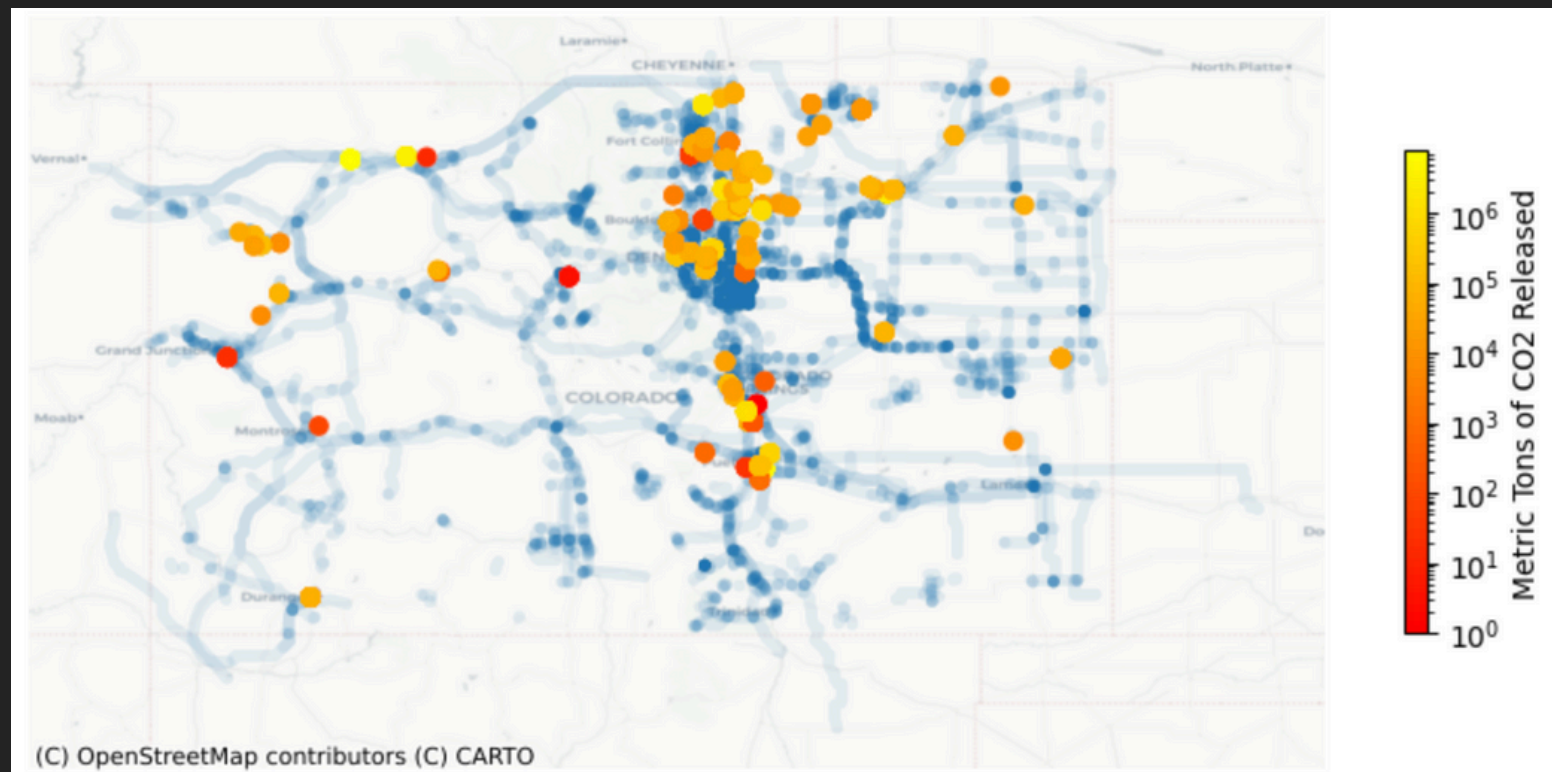
Creating Basic Maps

- Started with creating a qualitative map to show NERC Coordinators
- Shifted to working with quantitative data



Overlaying Data

- *Overlaying* data involves comparing two data sets and selecting data points based on common characteristics
- Made several new advancements using the Geopandas library to make overlay maps more intuitive and useful



Analysis

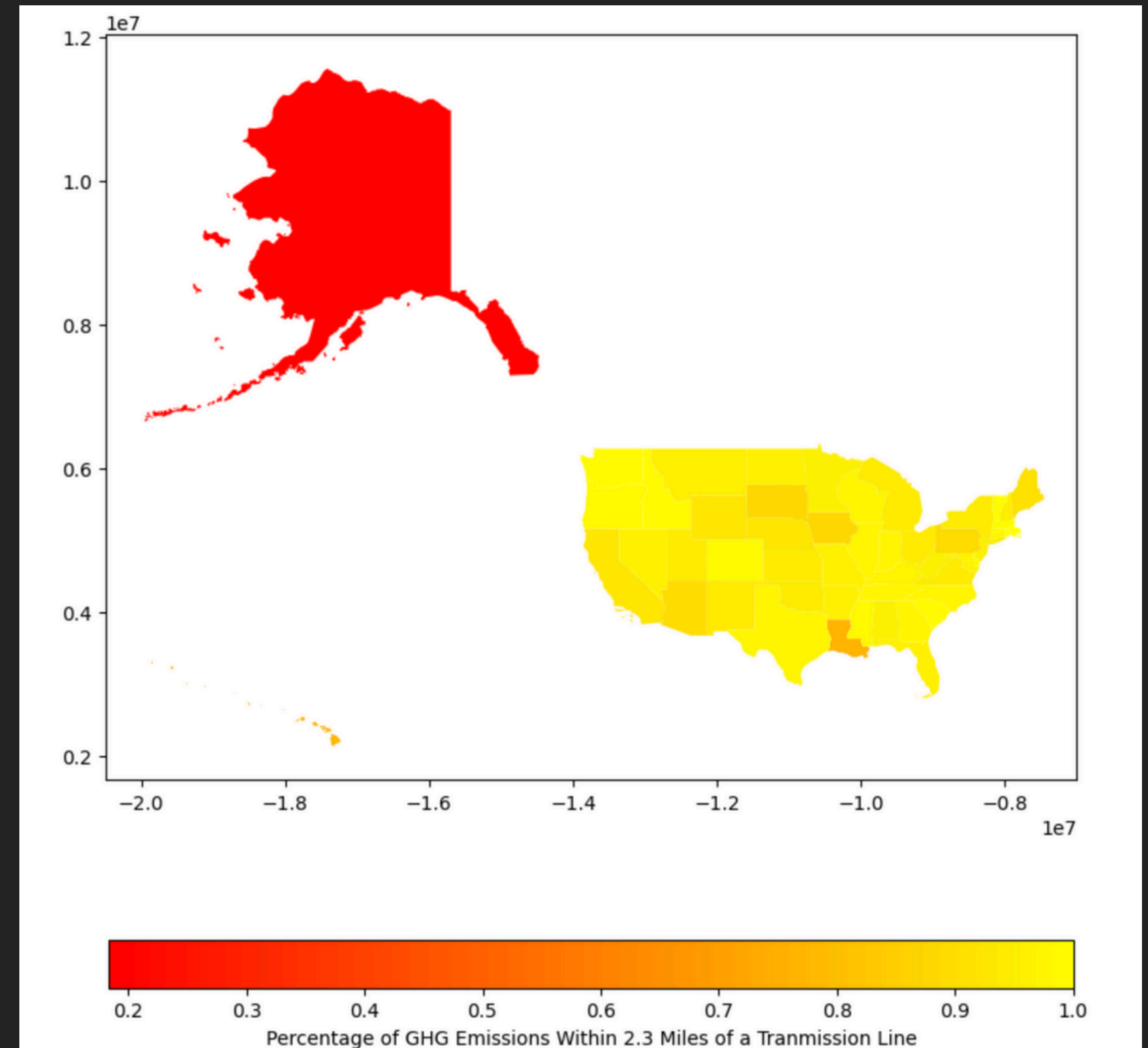
- Found GHG emission and power grid correlation
- Gives an insight into where emissions need to be reduced

```
... Total emissions inside 2.3 miles of the power grid: (Metric Tons of CO2)
2509019979
Total national CO2 emissions: (Metric Tons of CO2)
2677808848
Total emissions outside of 2.3 miles of the power grid: (Metric Tons of CO2)
168788869
Percentage of emissions within 2.3 miles of the power grid:
93.6967543771444

Total emission sources:
7508
Total emission sources within 2.3 miles of the grid:
6189
Percentage of emission sources within 2.3 miles of the grid
82.43207245604688
```

Further Mapping

- Graphed the percentage of emissions occurring within 2.3 miles of the power grid by state
- Found that most states had similar percentages of emissions within 2.3 miles of power grid - but there are several exceptions.



Significance

How can this project be used?

- Provides documentation to mapping technology
- Gives insight into where clean energy projects are needed
 - Shows where emissions are occurring

Further Information

- Github: <https://github.com/terranexum/cloud-ai/tree/main>
- Terranexum: <https://www.terranexum.com/>
- Geopandas: <https://geopandas.org/en/stable/>